

SC1007 Data Structures and Algorithms

Lab 5: Analysis of Algorithms

Practice Questions

- Q1** Given an array of n elements, try Bubble Sort and Merge Sort, and compare their execution times for different input sizes. You can generate an array of random numbers (e.g., size 1,000, 10,000, 100,000). Use the time module of Python to measure execution time. Plot the results of execution time for different input sizes. What is the time complexity for each algorithm?
- Q2** Try the recursive and iterative Fibonacci. Measure execution time for different n values (e.g., 20, 30, 40, 45). Plot the results of execution time for different input. What is the time complexity for each algorithm?